

The 4-node star system stability score regression

You are working for a smart grid company. Analysis of the local stability of the 4-node star system (the electricity producer is in the center) can be done using features extracted from the system. The lower the stability score is, the more stable the system is.

The dataset contains 12 features extracted from the system, and the stability score should be predicted in a regression problem setting.

The tasks include:

1. Train a neural network model using the training set ('TrainingSet.mat') and test it with the test set ('TestSet.mat') (see reading materials). The model should include an input layer with 'rescale-zero-one' normalization, two hidden layers with ReLU activation, an output layer, and a regression layer.
2. Use the Adam Optimizer and train the model on a CPU. Evaluate the regression with the root mean squared error.
3. Try to fine-tune the neural network structure for better performance. Part of the code has been provided, and you need to fill in the missing part based on the comments.

Script ?

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```
14 layers = [  
15     featureInputLayer(size(tblTrain,2)-1, "Normalization", "rescale-zero-one" )  
16     fullyConnectedLayer( 40 )           % hidden layer neurons: 40  
17     reluLayer           % ReLU activation func. (not a new layer of neurons)  
18  
19     % The other layer here, also 40 neurons and ReLU activation func.  
20     fullyConnectedLayer( 40 )           % hidden layer neurons: 40  
21     reluLayer           % ReLU activation func. (not a new layer of neurons)  
22  
23     fullyConnectedLayer(1) % Output layer: # output dim.  
24     regressionLayer];  
25  
26 options = trainingOptions('adam', ... % using adam optimizer  
27     'MaxEpochs', 1000 , ... % number of epochs of training, 1000  
28     'MiniBatchSize', size(tblTrain,1), ... % batch training, please explain.  
29     'L2Regularization', 1e-3, ... % L2 Regularization, please explain.  
30     'Shuffle' , 'every-epoch' % shuffling the training samples
```

Run Script ?

Tests: All Tests Passed Submit ?

✔ Data splitting

✔ Normalization

✔ Number of layers

✔ Adam optimizer

✔ Training on CPU

✔ Training accuracy

Output

Code ran without output.